

Mark schemes

Q1.

(a) $\frac{dy}{dx} = \frac{2}{\sqrt{x}}$

accept $2x^{-\frac{1}{2}}$ etc

B1

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{4}{x}}$$

ft sign error in $\frac{dy}{dx}$

$$= \sqrt{\frac{x+4}{x}}$$

AG

M1A1F

A1

4

(b) (i) $x = 4 \sinh^2 \theta, dx = 8 \sinh \theta \cosh \theta d\theta$

M1 for any attempt at $\frac{dy}{d\theta}$

M1A1

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$$= \int \frac{\sqrt{4 \sinh^2 \theta + 4}}{8 \sinh \theta \cosh \theta} d\theta$$

M1

$$= \int \frac{2 \cosh \theta}{2 \sinh \theta} 8 \sinh \theta \cosh \theta d\theta$$

ie use of $\cosh^2 \theta - \sinh^2 \theta = 1$

m1

$$= \int 8 \cosh^2 \theta d\theta$$

AG

A1

5

- (ii) Use of $2\cosh^2 \theta = 1 + \cosh 2\theta$
allow if sign error

M1

$$= \int 4(1 + \cosh 2\theta) d\theta$$

oe

A1

$$= 4\theta + 2 \sinh 2\theta$$

oe

A1F

Use of $\sinh 2\theta = 2\sinh \theta \cosh \theta$

m1

$$= 4 \sinh^{-1} \frac{1}{2} + 4 \times \frac{1}{2} \sqrt{1 + \frac{1}{4}}$$

A1F

$$= 4 \sinh^{-1} \frac{1}{2} + \sqrt{5}$$

AG

A1

6 [15]

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Examiner reports

Q1.

This question was generally answered well and many candidates were able to score 12 out of the available 15 marks. Part (a) was well answered apart from a few candidates

who wrote $\frac{dy}{dx} = 2x^{-\frac{1}{2}}$ followed by $\frac{1}{2x^{\frac{1}{2}}}$. In part (b) there were two main sources of

error. The first was to interchange dx with $d\theta$ without any consideration of $\frac{dx}{d\theta}$, and the

second was to write $\sqrt{\frac{4 \sinh^2 \theta + 4}{4 \sinh^2 \theta}}$ as $\frac{2 \sinh \theta + 2}{2 \sinh \theta}$. There were also a few candidates who were unable to differentiate $4 \sinh^2 \theta$.

In part (b)(i), most candidates were able to integrate $8 \cosh^2 \theta$ correctly but few were able to arrive at the printed result in part (b)(ii). Two factors contributed to this. Candidates either failed to change the limits for x to the corresponding limits for θ or else wrote the answer with no evident method. This was unacceptable as the answer for the arc lengths was given.

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